

Analytic cosmological derivatives of the Limber C_ℓ^{ss} and $\xi_\pm$ Chain rule on Python-side response tables (plan for `cosmo2D_fisher.c`)

Cocoa / CosmoLike — design note

2026-09-27 (C draft validated; cosmolike noise test)

Summary

Differentiating the Limber integral under the integral sign gives every cosmological derivative of $C_{ss}^{ij}(\ell)$ — and, by linearity, of $\xi_\pm^{ij}(\theta)$ — in closed form. The only inputs are four responses per parameter X , supplied from Python on the same grids as `set_cosmology`:

$$R_X(k, z) = \left. \frac{\partial \ln P_{\text{NL}}}{\partial X} \right|_{k,z}, \quad \chi_X(z) = \left. \frac{\partial \chi}{\partial X} \right|_z, \quad \left. \frac{\partial \ln G}{\partial X} \right|_z, \quad \epsilon_X = \frac{\partial \ln \Omega_m}{\partial X}.$$

Everything else — $H(z)$ and its response, the lensing-efficiency integral, the $k = \ell/\chi$ shift of P_{NL} , the measure, the NLA amplitude — is analytic, using cubic-spline derivatives of $\chi(z)$ and $\chi_X(z)$ and the bicubic-spline slope $\partial \ln P_{\text{NL}} / \partial \ln k$. **Verified** against central finite differences of the full C_ℓ for $\ln A_s$, Ω_m and w_0 (halofit, NLA): agreement $\leq 1.1 \times 10^{-4}$ at $30 \leq \ell \leq 3000$ (Table 1), the level of the finite-difference truncation itself. The w_0 case uses no parameter-specific code.

What cannot be avoided: $R_X(k, z)$. It is not contained in any single-cosmology $P(k, z)$ table; the best table-only models miss $d \ln C / d \ln A_s$ by up to 35% and $d \ln C / d \Omega_m$ by up to 27% (Sec. 5). So the Python side must provide it (e.g. central differences of the CAMB tables only). **Scope:** NLA intrinsic alignments; TATT is excluded (its one-loop kernels need the FAST-PT FFTs of the response, which would complicate cosmolike).

1 The integral cosmolike computes

In cosmolike units (χ in c/H_0 , k in H_0/c , both fixed in h -units), the Limber shear–shear spectrum for source bins i, j is an integral over the scale factor between parameter-independent limits (the source $n(z)$ support is fixed in redshift):

$$C^{ij}(\ell) = \Pi(\ell) \int_{a_{\min}}^{a_{\max}} da \underbrace{\frac{\chi'(a)}{\chi^2(a)}}_{w(a)} q_i(a) q_j(a) P_{\text{NL}}(k_\ell(a), z(a)), \quad k_\ell = \frac{\ell + \frac{1}{2}}{\chi(a)}, \quad (1)$$

with $\chi'(a) \equiv |d\chi/da| = \chi_z/a^2$ (cosmolike’s `dchida`) and $\Pi(\ell) = (\ell - 1)\ell(\ell + 1)(\ell + 2)/(\ell + \frac{1}{2})^4$ (1812.05995 eqs. 74–79). For NLA the E-mode kernel factorizes per bin ($C_{BB} = 0$):

$$q_b = W_{\kappa,b} - W_{s,b} C_1, \quad W_{\kappa,b} = \frac{3}{2} \Omega_m \frac{\chi}{a} g_b(a), \quad W_{s,b} = n_b(z) E(z), \quad (2)$$

$$g_b(a) = \mathcal{P}_b(a) - \chi(a) \mathcal{Q}_b(a), \quad \mathcal{P}_b = \int_{a_{\min}}^a \frac{da'}{a'^2} n_b, \quad \mathcal{Q}_b = \int_{a_{\min}}^a \frac{da'}{a'^2} \frac{n_b}{\chi(a')}, \quad (3)$$

$E(z) = H/H_0 = 1/\chi_z(z)$ with $\chi_z \equiv d\chi/dz$ (the cosmologic `h0`), and the NLA amplitude $C_1(z) = A(z) \Omega_m \bar{C}_1 \rho_{\text{cr}}/D(z)$, $D(0) = 1$.

2 Differentiating under the integral

Because a and the limits in Eq. (1) do not depend on the cosmology, ∂_X commutes with the integral. Differentiate at fixed a (equivalently fixed z):

$$\partial_X C^{ij}(\ell) = \Pi(\ell) \int da w P_{\text{NL}} \left[\Lambda_X(\ell, a) q_i q_j + (\partial_X q_i) q_j + q_i (\partial_X q_j) \right] \quad (4)$$

with $\Lambda_X = \partial_X \ln w + \partial_X \ln P_{\text{NL}}$. Every factor follows from the four inputs:

Measure. $w = \chi'/\chi^2$ and $\chi' = \chi_z/a^2$ at fixed a , so

$$\partial_X \ln w = \frac{\chi'_X(z)}{\chi_z(z)} - 2 \frac{\chi_X}{\chi}, \quad (5)$$

where $\chi'_X \equiv d\chi_X/dz$ is the *cubic-spline derivative* of the input $\chi_X(z)$.

Power spectrum. P_{NL} depends on X explicitly and through $k_\ell = \ell/\chi$:

$$\partial_X \ln P_{\text{NL}}(k_\ell(a), z) = -n_{\text{eff}}(k_\ell, z) \frac{\chi_X}{\chi} + R_X(k_\ell, z), \quad n_{\text{eff}} \equiv \frac{\partial \ln P_{\text{NL}}}{\partial \ln k} \quad (6)$$

(n_{eff} from the bicubic spline of $\ln P_{\text{NL}}$). The first term is the chain rule the spline provides; the second is the explicit response, which no spline of one table can provide (Sec. 5).

Expansion rate. $E = 1/\chi_z$ gives $\partial_X \ln E = -\chi'_X/\chi_z$ — again only the spline derivative of χ_X .

Lensing efficiency. $\mathcal{P}_b$ is cosmology-independent; $\partial_X \mathcal{Q}_b = -\int da' a'^{-2} n_b \chi_X/\chi^2$. Hence

$$\partial_X g_b = -\chi_X \mathcal{Q}_b + \chi \mathcal{R}_b, \quad \mathcal{R}_b(a) = \int_{a_{\text{min}}}^a \frac{da'}{a'^2} \frac{n_b(z') \chi_X(a')}{\chi^2(a')}, \quad (7)$$

one more cumulative trapezoid on the same fine grid `g_tomo` uses.

Kernels.

$$\partial_X W_{\kappa,b} = W_{\kappa,b} \left(\epsilon_X + \frac{\chi_X}{\chi} \right) + \frac{3}{2} \Omega_m \frac{\chi}{a} \partial_X g_b, \quad (8)$$

$$\partial_X W_{s,b} = -W_{s,b} \frac{\chi'_X}{\chi_z}, \quad (9)$$

$$\partial_X C_1 = C_1(\epsilon_X - \gamma_X), \quad \gamma_X(z) = \frac{\partial \ln G}{\partial X}(z) - \frac{\partial \ln G}{\partial X}(0), \quad (10)$$

the last because cosmologic's $D(z) = a G(z)/G(0)$ is normalized at $z = 0$. Then $\partial_X q_b = \partial_X W_{\kappa,b} - (\partial_X W_{s,b}) C_1 - W_{s,b} \partial_X C_1$.

Structure. Per quadrature node, wP , Λ_X , q_b and $\partial_X q_b$ are precomputed; the (ℓ, pair) fill of Eq. (4) is then the same branch-free SIMD reduction as `C_ss_tomo_limber_work`, computing C^{ij} and all $\partial_X C^{ij}$ in one pass.

3 Real space

cosmolike’s $\xi_{\pm}^{ij}(\theta) = \sum_{\ell} G_{\pm}(\ell, \theta) [C_{EE}^{ij}(\ell) \pm C_{BB}^{ij}(\ell)]$ uses bin-averaged Legendre kernels $G_{\pm}$ that do not depend on the cosmology, so the map is linear and exact:

$$\partial_X \xi_{\pm}^{ij}(\theta) = \sum_{\ell} G_{\pm}(\ell, \theta) \partial_X C_{EE}^{ij}(\ell), \quad \partial_X \ln \xi_{\pm} = \partial_X \xi_{\pm} / \xi_{\pm}. \quad (11)$$

In practice $\partial_X C$ is tabulated on the log- ℓ grid and gathered to every integer ℓ with `limber_fill_interp`, exactly as `xi_pm_tomo` does for C . Shear calibration $(1 + m_i)(1 + m_j)$ multiplies outside and cancels in $\partial_X \ln \xi_{\pm}$.

4 Verification

A Python implementation of Eqs. (1)–(4) (CAMB, Takahashi halofit, $m_{\nu} = 0.06 \text{ eV}$, two Gaussian source bins at $z = 0.5, 1.1$, NLA with $A_1 = 0.6$, $\eta = -1.5$) takes the four inputs as central differences of the CAMB tables on fixed grids — exactly what the Python side would hand to cosmolike. The truth is the central difference of the full C_{ℓ} , recomputing geometry, kernels, growth and P_{NL} at $X \pm h$ ($h = 0.01$ in $\ln A_s$, 0.005 in Ω_m , 0.02 in w_0).

Table 1: $d \ln C^{ij} / dX$: finite difference, and analytic/FD–1.

ℓ	bins	$X = \ln A_s$		$X = \Omega_m$		$X = w_0$	
		FD	rel. diff	FD	rel. diff	FD	rel. diff
30	(0,0)	1.02396	-2.4e-06	8.16100	-5.2e-05	-0.85127	-1.0e-05
30	(1,1)	1.00113	-5.2e-07	5.87218	-5.1e-05	-0.79806	-2.4e-05
30	(0,1)	1.01284	-1.4e-06	7.40933	-5.3e-05	-0.86004	-1.7e-05
100	(0,0)	1.19447	+9.5e-06	10.70446	-2.1e-05	-0.94115	-4.8e-05
100	(1,1)	1.06280	+6.6e-06	8.40663	-4.1e-05	-0.89853	-5.0e-05
100	(0,1)	1.14412	+9.4e-06	9.95070	-2.5e-05	-0.96658	-6.2e-05
300	(0,0)	1.45336	-2.3e-05	12.19212	-3.6e-05	-0.96586	-5.3e-05
300	(1,1)	1.27960	+3.3e-06	10.61055	-3.2e-05	-0.97054	-8.9e-05
300	(0,1)	1.41760	-1.8e-05	11.75446	-3.2e-05	-1.00069	-1.0e-04
1000	(0,0)	1.42906	-5.4e-05	12.15561	-7.3e-05	-0.92036	+6.9e-05
1000	(1,1)	1.48458	-4.3e-05	11.92446	-5.0e-05	-0.97633	-4.5e-05
1000	(0,1)	1.47246	-5.6e-05	12.13367	-6.9e-05	-0.97216	-8.9e-06
3000	(0,0)	1.14133	-4.0e-05	10.75940	-7.4e-05	-0.77947	+9.9e-05
3000	(1,1)	1.32769	-5.7e-05	11.48836	-5.4e-05	-0.88695	-3.7e-05
3000	(0,1)	1.19839	-3.8e-05	10.90386	-6.2e-05	-0.85035	+1.9e-05

5 Why R_X must be an input

For $\ln A_s$ the projection terms vanish identically (A_s does not touch geometry, kernels or growth normalization), so the entire derivative is the kernel-weighted average of R_{A_s} . We tested the two

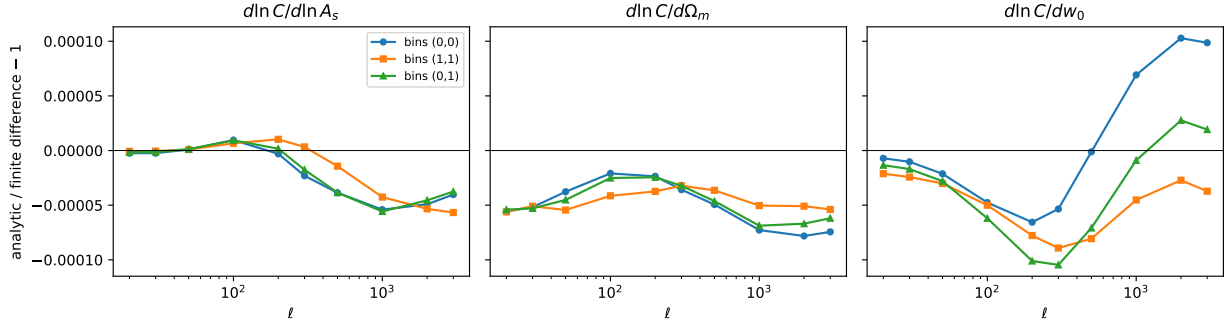


Figure 1: Analytic derivative over finite difference, minus one. The residual ($\lesssim 10^{-4}$) is the $O(h^2)$ truncation of the finite differences.

best ways to build R_X from the fiducial tables alone:

- *Amplitude–redshift degeneracy*: $R_{A_s} \approx \partial_z \ln P_{\text{NL}} / \partial_z \ln P_L$. Exact in linear theory, but halofit depends on redshift also through $\Omega_m(z)$ (e.g. the one-halo slope $y^{3f_1(\Omega_m(z))}$), which contaminates the z -derivative by ~ 0.3 in R at $k \gtrsim 1 h/\text{Mpc}$.
- *Shape shift for Ω_m* : $\Gamma = \Omega_m h$ scaling of the transfer function, amplitude via the degeneracy, growth from the sensitivity ODE.

Table 2: Table-only response models versus finite differences (bins (0, 0)).

ℓ	$d \ln C / d \ln A_s$			$d \ln C / d \Omega_m$		
	FD	model	error	FD	model	error
30	1.0240	1.0140	-1.0%	8.1610	7.1515	-12.4%
100	1.1945	1.1764	-1.5%	10.7045	9.1855	-14.2%
300	1.4534	1.4947	+2.8%	12.1921	11.2529	-7.7%
1000	1.4291	1.6257	+13.8%	12.1556	12.9572	+6.6%
3000	1.1413	1.5363	+34.6%	10.7594	13.6514	+26.9%

6 Cost and precision

A finite-difference Fisher needs $2N$ (five-point: $4N$) CAMB runs *and* $2N$ ($4N$) full cosmolike evaluations. The analytic route needs the same CAMB runs (to build R_X , χ_X , $\partial_X \ln G$) but only *one* cosmolike pass that returns C and all N derivatives together. Since CAMB dominates the wall time, the speed-up is on the cosmolike side (the C draft returns C plus three derivatives in ~ 40 ms, $\xi_{\pm}$ plus three derivatives in ~ 90 ms). On precision, the measurements of Sec. 7 overturn the initial expectation that cosmolike’s interpolants make finite-difference Fishers noisy: cosmolike adds no measurable noise, and the noise of the notebook Fisher is CAMB’s halofit (Sec. 8), which a one-line CAMB fix removes. The analytic route is therefore not needed for stability; it remains the natural adapter for differentiable (JAX) Boltzmann codes, which give the four inputs exactly by automatic differentiation.

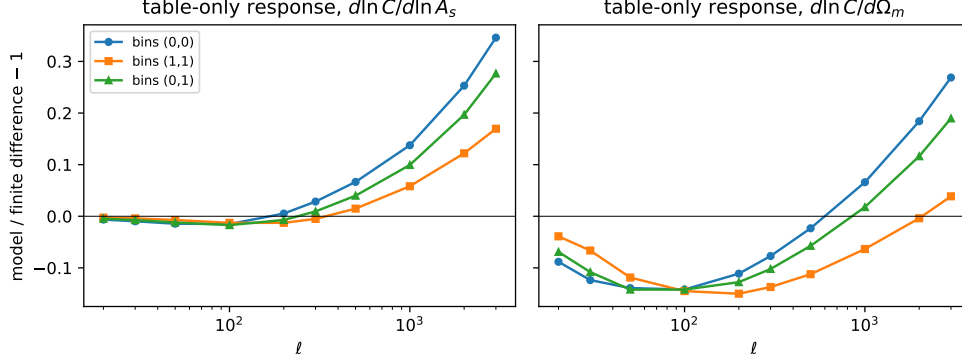


Figure 2: Table-only response models: errors of 10–35% at $\ell \geq 1000$.

7 Validation of the C draft

`cosmo2d_fisher.c` was built into a scratch copy of the `lsst_y1` interface (5 source bins, NLA) and checked two ways.

Chain rule, CAMB noise removed. The perturbed cosmologies are synthetic: every `set_cosmology` table is the fiducial one moved exactly along the loaded response ($\ln P_{\text{NL}} \pm s R_X$, $\chi \pm s \chi_X$, $\ln G \pm s \partial_X \ln G$, $\Omega_m(1 \pm s \epsilon_X)$), and `cosmolike`’s own C_ℓ and `xi_pm_tomo` are differenced along that path. Maximum $|\text{analytic}/\text{FD} - 1|$ over $30 \leq \ell \leq 3000$ and all 15 pairs, at $s = 10^{-4}$: 4×10^{-11} for $\ln A_s$, 5×10^{-5} for Ω_m , 4×10^{-4} for w_0 (step-independent: the analytic side uses spline derivatives, `cosmolike`’s C piecewise-linear interpolants); for $\xi_\pm$: 3×10^{-9} , 3×10^{-6} , 3×10^{-5} . The draft’s C and $\xi_\pm$ themselves equal the likelihood’s to 10^{-15} and 5×10^{-14} .

Against real CAMB finite differences. With the same step for the response and for the finite difference, the two agree to $\lesssim 6 \times 10^{-4}$ for $\ell \leq 1000$. Both drift together with the step (e.g. $d \ln C / d \ln A_s$ for bins (4, 4) at $\ell = 1000$: 1.24, 1.50, 1.40, 1.44 at $s = 0.0025$ –0.02), and one-sided differences disagree by up to $\sim 20\%$: the fiducial CAMB table differs from the mean of the $\pm$ tables by up to 5×10^{-3} in $\ln P_{\text{NL}}$. That jitter, not `cosmolike`, sets the precision floor.

8 Does cosmolike add noise to finite-difference Fishers?

Cosmolike alone. Along the smooth synthetic path above, finite differences of `cosmolike`’s C_ℓ and ξ_+ at Fisher-sized steps converge to the analytic derivative as s^2 (3-point) and s^4 (5-point stencil): for $\ln A_s$, 6.7×10^{-13} at $s = 10^{-3}$ and 2.1×10^{-6} at $s = 0.06$ with the 5-point stencil. Pure truncation, no noise.

CAMB. $|\ln P(X_0) - \frac{1}{2}[\ln P(X_0+s) + \ln P(X_0-s)]|$ reaches 5×10^{-3} for P_{NL} (but 10^{-14} for P_L) at every CAMB accuracy setting tried (`AccuracyBoost` 1–3, `CAMBAccuracyBoost` 2, `k_per_logint` 40). It is coherent over whole redshift slices: CAMB’s `halofit` stops the bisection for the nonlinear scale at $|\sigma(R) - 1| \leq 10^{-3}$ (`fortran/halofit.f90`, line 322), and the stopping point jumps with the cosmology. Tightening the tolerance to 10^{-7} (a scratch CAMB build) brings the jitter to 2.9×10^{-5} , the genuine curvature term, at no measurable cost.

The notebook Fisher. The `lsst_y1` notebook forecast (17 parameters, 5-point stencil, relative step h , FoM of (A_s, Ω_m)) reproduced at **AccuracyBoost 1**:

case	$h = 0.01$	0.02	0.03	0.05	0.08
stock CAMB (the notebook)	205.8	159.0	179.4	140.3	153.1
stock CAMB, CAMB boost 3	205.0	158.5	180.7	140.4	151.3
cosmolike only (synthetic inputs)	152.71	152.71	152.71	152.72	152.80
halofit tolerance 10^{-7}	146.0	145.8	145.6	145.6	147.1

With the patched CAMB, **AccuracyBoost 2** and **3** give 145.2 and 145.1 at $h = 0.03$: once the jitter is gone, the finite-difference Fisher is stable and converged at the default accuracy.

9 Scope and caveats

- **NLA only.** TATT one-loop kernels are FAST-PT convolutions of P_L ; their response would need FFTs inside cosmolike. Excluded by design; TATT Fisher stays on finite differences.
- **Parameter-agnostic, with the explicit Ω_m in ϵ_X .** Ω_m appears explicitly twice — the lensing prefactor $\frac{3}{2}\Omega_m(H_0/c)^2$ and the NLA amplitude $\Omega_m\bar{C}_1\rho_{\text{cr}}/D$ — and both are carried by $\epsilon_X = \partial \ln \Omega_m / \partial X$ at fixed other sampled parameters. In Cocoa’s basis $(\Omega_m, \Omega_b, H_0, A_s, n_s, m_\nu, w_0, w_a)$: $\epsilon_{\Omega_m} = 1/\Omega_m$ and $\epsilon_X = 0$ for all others — for h because Ω_m is itself sampled and cosmolike works in h -units (no explicit h in $(H_0/c)^2$ or $\bar{C}_1\rho_{\text{cr}}$), for m_ν because Cocoa’s Ω_m includes the neutrinos. A basis sampling ω_m instead would give $\epsilon_h = -2/h$. With that one scalar per parameter, the same code serves $A_s, \Omega_m, n_s, w_0, w_a, h, m_\nu$.
- **h and the k -grid.** The CAMB tables are shifted to h/Mpc after evaluation, so for $X = h$ the perturbed tables must be re-interpolated onto the fiducial grid before differencing; for every other parameter the grids coincide.
- **Grids.** Inputs share the fiducial `set_cosmology` grids; χ_X is passed in absolute form (not $\ln \chi$) so the $z = 0$ node is regular.
- **Baryon contamination** multiplies P_{NL} by a cosmology-independent ratio table, so R_X is unchanged.